

Retrieval-Grounded Interactive Report Generation for Cataract Surgery Videos

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ABSTRACT

Automated cataract surgery video analysis has the potential to support surgical training, workflow assessment, and postoperative review. While recent deep learning approaches have achieved strong performance in surgical phase recognition and instrument detection, most remain limited to isolated prediction tasks. We present a retrieval-grounded workflow analysis framework that integrates temporal phase recognition, surgical instrument localization, structured timeline generation, uncertainty-aware analysis, and interactive reporting within a unified system. The framework combines a ResNet18 feature extractor with a Multi-Stage Temporal Convolutional Network (MS-TCN) for workflow recognition and a YOLOv8n detector for surgical instrument localization. Model predictions, timestamps, confidence scores, representative frames, and detected instruments are fused into a structured timeline that supports evidence-grounded question answering and automated report generation through an interactive interface. Evaluated on the Cataract-1K dataset, the proposed framework achieved 95.58% phase recognition accuracy with a macro F1 score of 0.947, while the instrument detector achieved an mAP50 of 0.860. Qualitative results demonstrate transparent retrieval-grounded interaction that enables efficient surgical video exploration, workflow review, and educational support.

Keywords: cataract surgery, surgical workflow analysis, phase recognition, instrument detection, medical video analysis, temporal convolutional networks, computer-assisted intervention, retrieval-grounded reporting

1. INTRODUCTION

Automated analysis of cataract surgery videos has become an important research area for surgical training, workflow assessment, quality assurance, and postoperative review. Cataract remains one of the leading causes of vision impairment worldwide, motivating the development of intelligent systems for surgical workflow analysis.¹ Recent deep learning approaches have achieved strong performance in surgical phase recognition and instrument detection using publicly available datasets such as Cataract-101 and Cataract-1K.^{2,3} However, most existing systems remain limited to isolated prediction tasks and provide limited support for interactive exploration of surgical procedures. We propose a retrieval-grounded workflow analysis framework that combines temporal phase recognition, surgical instrument localization, structured timeline generation, uncertainty-aware analysis, and interactive reporting within a unified system. Rather than producing isolated predictions, the framework organizes workflow information into a searchable timeline that enables evidence-grounded question answering and transparent surgical video review.

2. METHODS

The proposed framework, shown in Fig. 1, consists of five major components. Cataract surgery videos first undergo frame extraction and preprocessing before being analyzed by a ResNet18 feature extractor⁴ followed by a Multi-Stage Temporal Convolutional Network (MS-TCN)^{5,6} for temporal workflow recognition. A parallel

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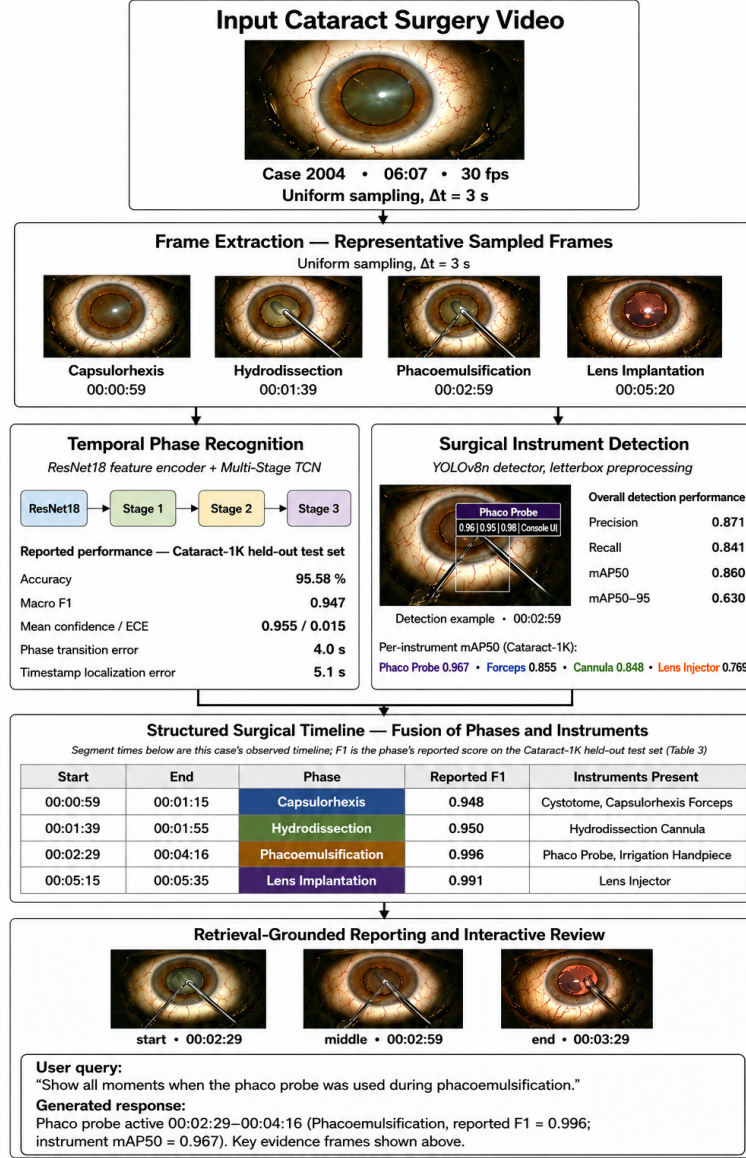


Figure 1. Overview of the proposed retrieval-grounded workflow analysis framework. The system combines temporal phase recognition, instrument detection, structured timeline generation, and evidence-grounded interactive reporting using real cataract surgery frames extracted from the evaluation dataset.

YOLOv8n detector⁷ localizes surgical instruments within each sampled frame. Predicted workflow phases, timestamps, confidence estimates, detected instruments, and representative evidence frames are fused into a structured surgical timeline. This timeline serves as the knowledge base for a retrieval engine inspired by retrieval-augmented generation (RAG),^{8,9} enabling evidence-grounded interaction through a natural-language chatbot. Rather than generating responses directly from model predictions, the chatbot retrieves relevant timeline segments and returns timestamps, supporting frames, detected instruments, confidence values, and uncertainty indicators associated with each query. As illustrated in Fig. 1, the framework integrates workflow recognition, instrument localization, structured timeline generation, and retrieval-grounded reporting into a unified pipeline for cataract surgery video analysis.

3. RESULTS

Experiments were conducted on the Cataract-1K dataset.³ The proposed ResNet18 + MS-TCN model substantially outperformed the CNN+GRU and frame-level ResNet18 baselines, achieving 95.58% accuracy and a macro F1 score of 0.947, as summarized in Table 1. The proposed instrument detector also achieved strong localization performance with an mAP50 of 0.860, as detailed in Table 2. Beyond quantitative evaluation, the proposed framework generates a structured surgical timeline that integrates workflow phases, confidence estimates, detected instruments, and review indicators into a unified chronological representation (Fig. 3). This timeline forms the foundation of the retrieval engine and enables interactive evidence-grounded exploration of surgical videos. As illustrated in Fig. 2, user queries retrieve the corresponding workflow phases, timestamps, representative frames, detected instruments, and confidence estimates directly from the structured timeline. This allows users to efficiently explore surgical procedures while ensuring that generated responses remain traceable to the underlying visual evidence.

Qualitative Retrieval Examples

Paper/demo examples: query → timestamps → frames/tools → grounded answer.

case_4750 - Show me phacoemulsification
← Query
^

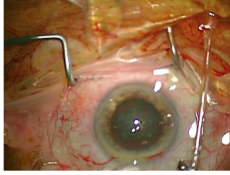
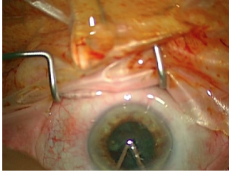
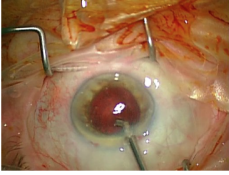
Retrieved timestamps: Phacoemulsification: 01:12-03:18 (phase, conf. 0.95), Phaco probe: 02:36 (tool, conf. 0.57), Key surgical phase: 01:12 (event, conf. 0.76)

← Response

Retrieved evidence ←

Phacoemulsification · 01:12-03:18 · confidence 0.95

Ultrasound energy breaks up the cloudy lens, and the fragments are removed by suction.

start · 01:12
middle · 02:12
end · 03:18

tool	timestamp	confidence	expected_in_phase	review_reason
Phaco probe	02:36	0.571	☒	None

Figure 2. Example retrieval-grounded interaction showing retrieved timestamps, representative frames, detected instruments, and evidence-grounded response generation.

Table 1. Workflow phase recognition performance on the Cataract-1K held-out test set.

Model	Accuracy (%)	Macro F1
CNN+GRU	77.57	0.746
ResNet18	87.85	0.857
ResNet18 + MS-TCN	95.58	0.947

Table 2. Summary of surgical instrument detection performance.

A) Overall Performance

Metric	Value
Precision	0.871
Recall	0.841
mAP50	0.860
mAP50-95	0.630

B) Per-Instrument Performance

Instrument	mAP50
Phaco Probe	0.967
Forceps	0.855
Cannula	0.848
Lens Injector	0.769

C) Tool Mode Ablation

Mode	Runtime (s)	Avg./Video
Off	14.25	0
Fast	19.84	19.67
Full	27.86	37.33

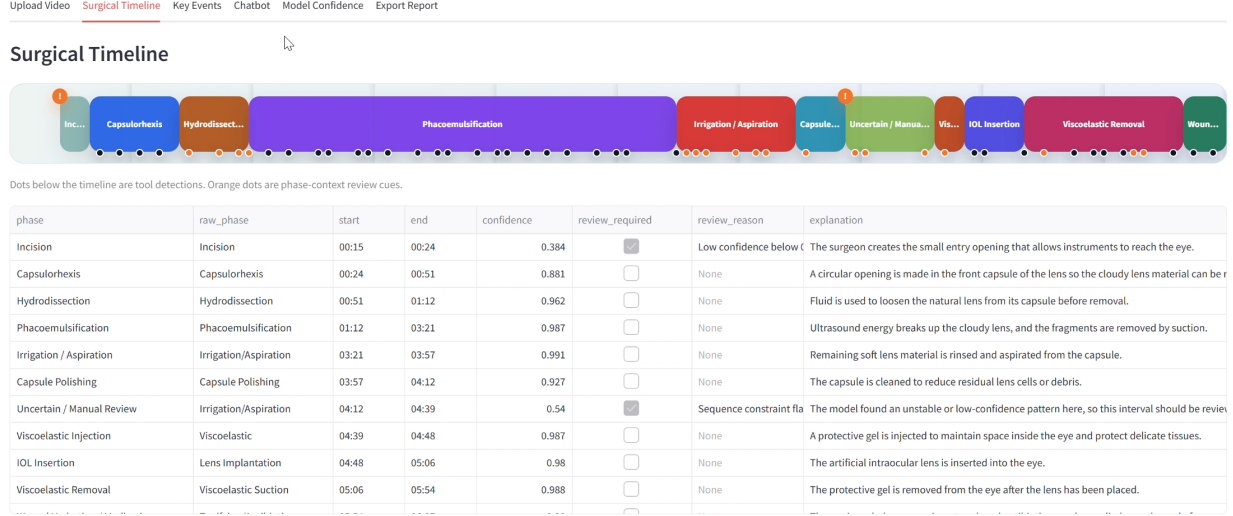


Figure 3. Example structured timeline generated by the proposed framework (Surgical Timeline tab). Workflow phases, confidence estimates, detected instruments, and review indicators are integrated into a unified representation for interactive surgical video review.

4. CONCLUSIONS

The proposed retrieval-grounded framework combines temporal workflow recognition, instrument localization, structured timeline generation, and evidence-grounded reporting within a unified cataract surgery video analysis system. Experimental results demonstrate accurate workflow recognition and reliable instrument localization while enabling transparent interaction through retrieval-grounded question answering. The framework represents a practical step toward interactive AI-assisted surgical video review and educational support.

Novelty

Unlike existing cataract surgery analysis systems that terminate at workflow prediction or instrument detection, the proposed framework integrates workflow recognition, instrument localization, structured timeline generation, uncertainty-aware analysis, and retrieval-grounded interactive reporting within a single platform. This allows users to explore surgical videos through evidence-grounded natural-language queries while maintaining direct traceability to the underlying visual evidence.

Disclosure

This work has not been previously published and is not under consideration elsewhere. An extended journal version with additional experiments and cross-dataset evaluation is planned.

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